



Justice perceptions of performance appraisal practices

Perceptions of
performance
appraisal

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Abstract

Purpose – The purpose of this paper is to explore the underlying structure of employees' justice perceptions in the context of their organizations' performance appraisal practices.

Design/methodology/approach – Ten multi-item scales were designed to measure the perceived fairness of performance appraisal practices. A nested confirmatory factor analysis of employee responses ($n = 188$) compared the four justice dimensions (i.e. procedural, distributive, interpersonal, informational) to five plausible alternatives. Construct validity was demonstrated through a structural equation model of matched employee and supervisor responses ($n = 117$).

Findings – The confirmatory factor analysis showed evidence of four distinct but highly correlated justice constructs. Results supported hypothesized relationships between procedural justice and helpful behaviors toward the organization via appraisal system satisfaction; distributive justice with appraisal satisfaction; and interpersonal and informational justice and helpful behaviors toward the supervisor via supervisor satisfaction.

Practical implications – This study underscores the importance of fostering perceptions of justice in the context of performance appraisal. The scales developed in this study could be used to isolate potential problems with an organization's performance appraisal practices.

Originality/value – The paper integrates prior research concerning the positive effects of procedural, distributive, interpersonal, and informational justice on affective and behavioral responses towards performance appraisals.

Keywords Performance appraisal, Job satisfaction, Psychology, Individual perception, Strategic objectives

Paper type Research paper

Introduction

An organization's performance appraisal system can be a practical tool for employee motivation and development when employees perceive their performance appraisals as accurate and fair (Ilgen *et al.*, 1979). Appraisal practices often include formal review and feedback sessions, and may include procedures for establishing work objectives, conducting self-appraisals, and setting performance goals. The processes inherent in these systems and the performance appraisal outcomes themselves can have an important influence on employees' reactions toward their work, their supervisors, and



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their organization as a whole. The appraisal process can also become a source of frustration and extreme dissatisfaction when employees perceive that the appraisal system is biased, political or irrelevant (Skarlicki and Folger, 1997).

Leaders of organizations may know that employees perceive their performance appraisal systems as unfair, but they have not had a convenient way of measuring their specific appraisal practices. Leaders who do not know the specific faults of current appraisal practices often assume that the entire system is bad. They may be limited to the choice of accepting the status quo, or scrapping old systems for new ones with the hope of improving employee reactions. New performance appraisal systems replace old, without any determination of the root causes of the dissatisfaction and without any basis for the new system. One possible way to rectify this situation is to provide leaders with the information necessary to make sensible decisions concerning their existing performance appraisal systems.

Organizational justice theory may provide a conceptual framework to collect this information. If justice perceptions are important to employees, then these perceptions should be related to attitudinal and behavioral reactions beyond the effects of the initial discrepancy between expected and actual performance ratings. Employees do not enjoy receiving a poor performance appraisal, but if they perceive that procedures and social interactions are fair, then discrepancies will be less likely to influence their attitudes and behaviors toward their supervisors and their organizations. The specific objectives of this research were twofold:

- (1) develop and confirm the component structure for a set of multi-item scales based on the various conceptualizations of justice in the performance appraisal context; and
- (2) provide construct validity evidence for the multifaceted performance appraisal justice measure as part of its larger nomological network.

Problems with performance appraisal systems and appraisal research

Folger *et al.* (1992, p. 130) suggest that researchers have not been able to substantially change employees' affective reactions to performance appraisal systems because their efforts have been embedded in an overly rational conceptualization. They state that traditional research has viewed performance appraisals as "analogous to the psychometric process of constructing a test". The test metaphor relies on the assumptions that an objective view of reality exists and, in the ideal appraisal situation, that both rater and the employee share this view. A vast amount of research has concentrated on methods to improve the reliability and validity of performance appraisal systems (Bernardin and Villanova, 1986). Researchers have investigated alternative rating formats, controls for rater bias, and various methods of rater training, but have had only limited success. Folger *et al.* suggest the problem with the test metaphor for performance appraisals is that the underlying assumptions are rarely true. Basic assumptions of the test approach like reliable and valid measurement, and an overly rational judgment process are inconsistent with the nature of work and the nature of managerial decision making (March, 1994).

An alternative view on performance appraisals comes from a political perspective (Longenecker *et al.*, 1987; Patz, 1975; Pfeffer, 1981; Tziner *et al.*, 1996). A survey by Bernardin and Villanova (1986) found that raters, ratees and administrators of performance appraisal systems believed that inaccuracies of ratings were due to the

deliberate distortion of performance ratings by the raters themselves. The political model suggests that performance appraisals occur in the context of raters' desires to project a favorable self image, obtain valuable outcomes for their units, portray themselves as caring individuals, and avoid negative consequences and confrontations (Cleveland and Murphy, 1992). The political model provides a compelling argument for the levels of inaccuracy that can be found in performance appraisals in spite of the extensive efforts made by researchers and practitioners over the years. The political metaphor fails, however, in much the same way as the test metaphor that preceded it. Performance appraisal processes may not be objective, but they are certainly not illusory. The political model ignores the checks and balances placed on the participants by the organization and the people within that organization. Formal processes, social norms, and ethical and legal standards constrain the instrumental goals of the person conducting the performance appraisal. With these constraints, the political model can provide only a partial explanation of employees' perceptions that performance appraisals are unfair.

Folger *et al.* (1992, p. 140) offer a "due process" model based on perceptions of procedural fairness as the key to help close the "fundamental gap between performance appraisal theory as explained by the traditional or rational approaches and performance appraisal practice as described by the political" model. The assumptions of a fairness model are less extreme than the assumptions of either the traditional rational model or of the political model. Accuracy does not require a shared objective reality between rater and ratee, but rather a shared view of acceptable standards and types of information that can be brought to bear as evidence to compare performance to those standards. Similarly fairness does not require a complete accounting of work output across all individuals, but rather requires a shared sense of how people should be treated, how rewards should be allocated, and how decisions should be made and explained. The due process model, although a step in the right direction, limits itself to structurally determined aspects of the appraisal system. It does not consider many of the social aspects of performance appraisal practices that are also important to perceptions of fairness. The same model also ignores fairness perceptions which are directly associated with performance appraisal outcomes. In order to incorporate these aspects into a more comprehensive model of performance appraisal practices, the interaction between outcomes and the procedures that lead to these outcomes, as well as the relative influences of social and structural forces needs to be understood.

A complete understanding of the effects of performance appraisal on individuals' perceptions and actions must depend on all aspects of the appraisal process. Research has revealed much about justice perceptions in the context of performance appraisal. Performance appraisal is more than the observation, judgment, evaluation, interviews, and formal documentation emphasized by the traditional rational model. Improvements cannot be limited to the formats, criteria, training, goal setting, feedback, and other methods by which researchers have attempted to improve the traditional model. Performance appraisal systems are also more than the personalities, self-interests, power and negotiations among their participants as suggested by the political model. Performance appraisal research must include all aspects of the performance appraisal system in an integrated framework. Such a framework needs to combine the social interactions among the people involved with the structural forces in the environment. These in combination shape the perceptions about the processes and

outcomes of performance appraisals. Researchers will only be able to explain performance appraisal phenomenon and organizations will only be able to improve their performance appraisal practices when they can understand the flaws in the system as a whole.

Organizational justice approach to performance appraisal research

The organizational justice literature provides a robust framework for explaining and improving perceptions about performance appraisals. Organizational justice is deeply rooted in social exchange theory. Social exchange theories make two basic assumptions about human behavior (Mowday, 1991): social relationships are viewed as exchange processes in which people make contributions for which they expect certain outcomes; and, individuals evaluate the fairness of these exchanges using information gained through social interactions.

The original version of social justice theory suggested that social exchanges were perceived as fair when people sensed that their contributions were in balance with their rewards (Adams, 1963; Homans, 1961). This equity theory later became known as the distributive form of organizational justice because it involved the allocation or distribution of outcomes (Greenberg, 1990). Subsequent research discovered that individuals would accept a certain amount of injustice in outcome distributions as long as they perceived that the procedures that led up to those outcomes were fair (Cropanzano and Folger, 1991; Greenberg, 1990; Leventhal, 1980). Procedural justice describes the phenomena of perceived fairness in the allocation process. Leventhal (1980) identified seven procedural categories that individuals can use in order to determine the fairness of organizational processes. These include procedures for selecting agents, setting ground rules, collecting information, making decisions, appealing decisions, safeguarding employee rights, and changing procedures. An individual's awareness of unfair practices in any one of the seven factors can lead to perceptions of injustice. Since the publication of Leventhal's model, researchers have clearly demonstrated the existence of two justice factors: a distributive factor associated with the fairness of distribution of outcomes, and a procedural factor associated with the fairness of the means used to determine the outcomes.

A third form of justice, introduced by Bies and his colleagues, focuses on the quality of interactions among people in the work environment (Bies and Moag, 1986; Bies and Shapiro, 1987). Bies argued that the quality of interpersonal treatment received during the enactment of organizational processes and distribution of organizational outcomes is an important contributor to fairness perceptions. In the past, there has been some disagreement among justice researchers as to whether interactional justice is distinct from procedural and distributive justice, or as an interpersonal aspect of procedural justice (Greenberg, 1993). However, Cohen-Charash (2001) meta-analysis offers support for three distinct justice dimensions (distributive, procedural, and interactional).

Greenberg (1993) argued that interactional justice actually consisted of two components. The first component refers to interpersonal justice, or the quality of the treatment that the target receives, and the second component refers to informational justice, or the procedural explanations for why something occurred (Colquitt *et al.*, 2001). Colquitt (2001) found that distributive, procedural, interpersonal and informational justice were distinct dimensions of organizational justice. Colquitt *et al.*'s (2001) meta-analysis also found evidence for four distinct justice constructs.

Interpersonal and informational justice were highly correlated, but not so highly correlated to combine them into one overall interactional justice measure. In addition, studies have found significant unique effects for interpersonal and informational justice (Colquitt, 2001; Kernan and Hanges, 2002).

Applying organizational justice theory to performance appraisal practices

More recently, Roch and Shanock (2006) used exchange theory to incorporate all four justice dimensions into one theoretical framework. They found that procedural, interactional, interpersonal, and informational justice were related to social relationships, either with the organization (i.e. procedural justice) or with the supervisor (i.e. interactional, interpersonal, and informational justice), whereas distributive justice is related more to an economic exchange relationship. In the current study, we draw upon this integrative framework and apply it specifically to a performance appraisal context. This conceptualization may hold the key to explaining employees' perceptions of fairness concerning their performance appraisals and appraisal systems. Below we discuss relevant performance appraisal literature pertaining to each of the four justice dimensions.

Procedural justice perceptions. According to Leventhal's (1980) model, judgments will depend on the relative weighting of the perceived fairness of the structural components of the performance appraisal procedure. Three specific procedures have shown prominence in the performance appraisal research (assigning raters, setting criteria and seeking appeals). Landy *et al.* (1978), Klasson *et al.* (1980), and Tang and Sarsfield-Baldwin (1996) found evidence that the assignment of raters influences perceptions of fairness and accuracy in performance appraisals. Folger *et al.* (1992) and the subsequent empirical work by Taylor *et al.* (1995) emphasized the importance of setting criteria and seeking appeals. Silverman and Wexley (1984) found that participation in construction of behaviorally anchored rating scales led to favorable perceptions regarding the performance appraisal interview process and outcomes. Stratton (1988), and Alexander and Ruderman (1987) found that perceptions of appeal procedures were positively related to evaluations of supervisors, trust in management, and job satisfaction.

Distributive justice perceptions. Distributive justice is deeply rooted in the research of the original equity theorists (Adams, 1963; Homans, 1961). There are two types of structural forces associated with the distributive justice of a performance appraisal as an outcome. The first type is decision norms (e.g. equity). Receivers of distributions structured to conform to existing social norms, like equity, typically believe that the distributions are fair. Raters, however, may also feel driven to develop appraisals that conform to other distribution norms such as equality, need, or social status which may seem unfair to those being rated (Leventhal, 1980). The second type of structural force relates to the personal goals of the rater (e.g. to motivate, teach, avoid conflict or gain personal favor). Whether employees consider a particular appraisal as fair or unfair can depend on their perceptions of the rater's goals. Employees may consider an appraisal as fair if they perceive that the rater is trying to motivate them, improve their performance or expand their perceptions of their own capabilities. Goals that may not be perceived as fair can include conflict avoidance, favoritism and politics.

Interpersonal justice perceptions. Interpersonal justice concerns fairness perceptions that relate to the way the rater treats the person being evaluated. Greenberg (1986a)

provided evidence that individuals are highly influenced by the sensitivity they are shown by their supervisors and other representatives within the organization. This is especially true when raters show concern for individuals regarding the outcomes they receive. Specifically, Greenberg found that apologies and other expressions of remorse by raters have been shown to mitigate ratees' perceptions of unfairness.

Informational justice perceptions. Informational justice concerns fairness perceptions based on the clarification of performance expectations and standards, feedback received, and explanation and justification of decisions. Like procedural justice, the focus is on the events which precede the determination of the outcome, but for informational justice, the perceptions are socially rather than structurally determined. Information about procedures can take the form of honest, sincere and logical explanations and justifications of any component of the allocation process. In the context of performance appraisals the most common interactions will involve the setting of performance goals and standards, routine feedback, and explanations during the performance appraisal interview.

The first objective of this study was to develop a set of measures based on the various conceptualizations of justice in the performance appraisal context and then confirm the component structure of those measures. Ten performance appraisal perceptual constructs (assigning raters, setting criteria, seeking appeals, ratings based on equity, absence of politics, respect, sensitivity, clarifying expectations, providing feedback, and explaining decisions) were grouped according to four justice dimensions (Colquitt, 2001; Roch and Shanock, 2006), as well as competing three and two-factor models (e.g. Moorman, 1991 and Skarlicki and Folger, 1997). Distinguishing among justice dimensions provides only partial evidence for the validity of the measures of the four constructs. Further evidence is required to show that the operational definitions of the constructs correlate with other constructs in predictable ways.

Consequences of perceived fairness of performance appraisal practices

The second objective of this study was to validate the multifaceted performance appraisal justice measure as part of its larger nomological network. The adopted model of perceptual, affective and behavioral constructs is consistent with organizational adaptation models such as that proposed by Hulin *et al.* (1985) and Organ (1995). Individuals' perceptions are related to their affective reactions, and their reactions are related to their behaviors. Employees form beliefs about their contributions to their organizations. These beliefs form in a context of self-perceptions of capabilities and efforts, as well as the perceptions of the capabilities and efforts of others. Upon receipt of a performance appraisal, perceptions of the appraisal may contradict their beliefs about their contributions. Unmet expectations stemming from these beliefs form a discrepancy that can lead to dissatisfaction. Employees also make logical connections between events and form beliefs about their environment and the people with whom they interact. The four categories of justice judgments may provide a possible framework for those logical connections. The affective component of the model identifies three possible attitudinal responses to these perceptions (satisfaction with the performance appraisal itself, appraisal system and the supervisor). The model also includes two behavioral constructs in the form of supervisor reports of citizenship behaviors toward the organization and toward the supervisor.

One way to describe the relationships between these constructs is through a discrepancy approach. A discrepancy can occur when a performance appraisal does not meet a person's beliefs about the rating that he or she should receive. Employees who have high opinions about their own work performance have a greater opportunity to be disappointed when they receive their performance appraisal than their peers with lower self-appraisals. This discrepancy leads to a global perception that the appraisal is inaccurate, unfair, or some combination of the two. This dissatisfaction will likely be focused on the performance appraisal itself, but may also be associated with the performance appraisal system or the supervisor. According to Hulin *et al.* (1985) the dissatisfaction will then lead to withdrawal behaviors that help ease the tension associated with the discrepancy. Organ (1995) suggests that withdrawal behaviors may include not being a good citizen in the organization. The discrepancy model is consistent with distributive justice theory (Adams, 1963; Homans, 1961) and many models of job attitude formation (see Hulin, 1991) and suggests that the difference between expected and received outcomes is the driving force for attitudes and behaviors in the performance appraisal context. If this model proved true, the prescriptive challenge would be to make both the expectations about ratings and the ratings themselves as accurate as possible. A task, as Folger *et al.* (1992) noted, that is doomed because it assumes an overly rational conceptualization of the performance appraisal process.

An alternative way to describe the relationship between these constructs is through a justice model. Consistent with referent cognition theory (Cropanzano and Folger, 1989), people will likely recalculate the distributive equity of performance appraisals based on their perceptions of the quality of the process and social interactions that led to the appraisal. Employees may accept a disappointing evaluation as fair and accurate if they decide that the appraisal process and outcomes were just, and that throughout the process the interpersonal and informational interactions were fair. Employees can then change their beliefs about their prior performance to be aligned with the received appraisal. An employee can also use the information to take appropriate action (e.g. improve performance) in order to change the perception of others to be in line with his or her desired self-perception. On the other hand, if any of the appraisal practices can be faulted as unfair, then resentment with the appraisal, the system and the supervisor will likely result. Resentment with the procedural, informational or interpersonal determinants then becomes the reason for dissatisfaction and a motivator for behavior. In the justice model, the discrepancy creates an initial dissatisfaction with the performance appraisal, but the affective reactions toward the appraisal system and the supervisor provide the driving influence for action.

The present study tests the competing discrepancy and justice models in order to determine whether justice perceptions are useful in explaining employee reactions in a performance appraisal context. Support for the justice model will provide evidence for Folger *et al.*'s (1992) position that justice perceptions play an important role in attitudes toward performance appraisals and performance appraisal systems. The comparison of the competing models was guided by the following hypothesis.

- H1.* Satisfaction with the appraisal system, appraisal and supervisor will be positively related to perceptions of performance appraisal practices beyond the effects of any discrepancy between beliefs of their contributions and received appraisal.

The justice model also indicates that employee attitudes will be related to their assessments of the various types of performance appraisal practices in meaningful ways. To the extent that individuals find the performance appraisal procedures, decisions, and interactions to be fair, their attitudes toward the appraisal, appraisal system and rater will likely be positive. Faults in the performance appraisal practices will likely be related to increased employee frustration and dissatisfaction with their appraisal system, rater and appraisal. Distinguishing different relationships between the justice dimensions and their consequences will provide additional evidence for the validity of the measures of the four constructs.

According to Sweeney and McFarlin's (1993) two-factor model, distributive justice is more closely related to "specific, person-referenced outcomes such as satisfaction with a pay raise or performance evaluation" whereas procedural justice is more closely related to "general evaluations of systems" (Colquitt *et al.*, 2001, p. 428). On the other hand, the agent-system model (Bies and Moag, 1986) asserts that both interpersonal and informational justice will be more strongly related to agent-referenced outcomes rather system-referenced outcomes. Based on both the two-factor model and agent-system model, the following hypotheses were tested:

- H2. Perceptions of distributive justice will be positively related to employees' satisfaction with the performance appraisal.
- H3. Perceptions of procedural justice will be positively related to employees' satisfaction with the performance appraisal system.
- H4. Perceptions of both interpersonal and informational justice will be positively related to employees' satisfaction with the supervisor.

Justice perceptions have also been shown to correlate with variables such as organizational citizenship behaviors (OCBs) (Folger *et al.*, 1992; Moorman, 1991). Masterson *et al.* (2000) found that procedural justice was related to organization directed OCBs via perceived organizational support whereas interactional justice was related to supervisor directed OCBs via leader member exchange. These findings suggest that it is important to match the source of justice with the outcome variable. This fits with Roch and Shanock (2006, p. 316), who found that "procedural justice is more closely associated with attitudes regarding social exchange with the organization and interactional justice is more closely associated with attitudes regarding social exchange with the supervisor". Similarly, Colquitt *et al.*'s (2001) meta-analysis found that both interpersonal and informational justice dimensions were more strongly related to individual-referenced OCBs. Through affective reactions, it is expected that the justice dimensions will have a positive relationship with the organizational outcome variables represented by the behavioral aspects of the model.

- H5. A positive relationship between procedural justice and citizenship behaviors toward the organization will be mediated through satisfaction with the appraisal system.
- H6. A positive relationship between interpersonal and informational justice perceptions and citizenship behaviors toward the supervisor will be mediated through satisfaction with the supervisor.

Method

A phased process patterned after the performance appraisal research conducted by Greenberg (1986b) and Tziner *et al.* (1996), and the construct validation study of Paullay *et al.* (1994) was followed to develop and provide evidence of construct validity for the set of multi-item scales designed to measure employee' perceptions of performance appraisal practices. The purposes of this process were to ensure the content validity and reliability of the scales developed in the study (phase 1), confirm the underlying structure of the justice scales (phase 2), and demonstrate the construct validity of the scales in an organizational setting (phase 3). The methodology for all three phases is described below.

Phase 1: Instrument development

The content development process capitalized on theoretical conceptualizations in the organizational justice and performance appraisal literature (e.g. Adams, 1963; Folger *et al.*, 1992; Leventhal, 1980), empirical research on justice perceptions and effectiveness in the performance appraisal context (e.g. Greenberg, 1986b; Landy *et al.*, 1978; Taylor *et al.*, 1995), legal considerations in performance appraisal cases (e.g. Bernardin and Beatty, 1984), and finally, observations from personal experiences as a rater and ratee. Items were created in accordance with the guidelines described in Nunnally and Bernstein (1974). Content validity, item clarity, and conceptual distinction among the four justice dimensions were ensured through a confirmatory categorization analysis (Paullay *et al.*, 1994). Five subject matter experts sorted the randomly ordered items according to descriptions of the constructs. The experts were chosen based on their experience as both a rater and ratee in the targeted organizations. One expert had rated at least ten subordinates, two had rated at least 20 subordinates, and the other two had rated more than 50 subordinates. Two raters had received between ten and 20 performance appraisals, and the other three had received more than 20 appraisals each during their career. Each subject matter expert received a packet of materials that included a cover letter describing the sorting task, envelopes with descriptions for each construct, and the items typed on individual slips of paper. The subject matter experts sorted the pool of randomly ordered items based on the construct descriptions. The experts were offered the opportunity to write additional items and make changes to items within the pool. Two of the five experts sorted the items in exact conformance with the assumed theoretical structure. The other experts had conformance rates of 88, 90 and 95 percent. Inter-rater reliability was calculated for the 50 items: 39 had 100 percent agreement, 11 had 80 percent agreement and two had 60 percent agreement. The two items that were incorrectly sorted by two experts were subsequently rewritten to improve clarification. The experts had no additional comments on item wording or missing content.

A pilot survey was administered to 45 United States Air Force officers to assess the reliability of each of the ten justice scales as well as the performance discrepancy measures required for subsequent phases. These officers were graduate students majoring in logistics, acquisition or information resource management. The students completed the survey based on their most recent performance appraisal from their last duty assignment. All 45 students had received a performance appraisal in the previous six months, and had access to their actual performance ratings. The officers shared a common performance evaluation system, but differed in terms of organization, duty

location and supervisor. A pilot survey was created by randomly sorting the items from the first phase. Respondents were instructed to indicate the extent that they perceived the content of each item described the performance appraisal practices of their organization. Respondents recorded their perceptions on a seven-point Likert type scale, ranging from "Strongly disagree" (1) to "Strongly agree" (7). Scale reliability was estimated by calculating the internal consistency of each multi-item scale as indexed by Cronbach's coefficient alpha (α).

The reliability of nine scales exceeded the normal acceptable limits of $\alpha = 0.70$ or research designed to make decisions affecting groups (Nunnally and Bernstein, 1974). The scale measuring absence of political goals approached the minimal acceptable level of internal consistency ($\alpha = 0.68$). This measure was retained for subsequent phases to ensure that there were two independent measures for the distributive justice construct. Perceptions of performance appraisal practices, on average, were greater than 4.0 (the neutral point of the scale) for all ten scales. The most favorable responses concerned the interpersonal justice scales (respect and sincerity shown by the supervisor). The least favorable perceptions concerned appraisal practices for setting criteria, with scores just above the neutral mark. Items and reliability estimates for the ten justice scales appear in Table I.

The pilot survey also contained items designed to measure the perceived discrepancy between employees' beliefs about the quality of their performance, and their beliefs about their most recent performance appraisal. Two constructs, perceptions of work performance and perceptions of performance rating captured the potential discrepancy observed between individuals' expectations and the actual performance ratings for the previous performance period. Work performance perceptions were measured using three items. Respondents rated their work performance during the last rating period on a seven-point scale ranging from "unsuccessful" (1) to "truly outstanding" (7). Respondents also indicated their beliefs about their work quality on a seven-point scale ranging from "well below acceptable standards" (1) to "well above acceptable standards" (7). The third item asked the respondents to indicate their beliefs about their contributions to their unit during the last rating period. Responses to the third item were marked on a seven-point scale ranging from "well below what was expected for my position" (1) to "well above what was expected for my position" (7). The measure of performance ratings perceptions used the same items and anchors, but focused on perceptions of what their most recent performance rating indicated about their work performance, work quality, and contributions. The reliability estimates for the work performance and performance appraisal perception scales were 0.88 and 0.87, respectively. Overall, the officers had scores very close to the maximum 21 points on the perceptions of performance appraisal scale ($M = 19.09$, $SD = 1.82$) and on the perceptions of work performance scale ($M = 17.96$, $SD = 2.15$), indicating very positive ratings and beliefs about their performance. A performance discrepancy perceptions measure was created by subtracting item scores on the work performance scale from the corresponding scores on the performance appraisal scale. This formulation is consistent with Tisak and Smith's (1994, p. 675) definition of difference scores "as the difference between distinct but conceptually linked constructs". A positive discrepancy implies the performance appraisal exceeded the person's beliefs about his or her performance. A negative discrepancy implies that the appraisal fell short of the employee's beliefs. The

Assigning raters ($\alpha = 0.90, 0.95$)	I am assigned a rater who is <i>qualified to evaluate my work, understands the requirements and constraints of my work, and is familiar with the rating formats and procedures</i> . Procedures ensure my rater who knows <i>what I am supposed to be doing and how to evaluate my performance</i> .
Setting criteria ($\alpha = 0.80, 0.90$)	<i>My organization requires that standards be set for me before the start of a reporting period</i> . Procedures make sure that performance standards <i>measure what I really do for the organization and are stable over time; and procedures allow me to help set the standards used to evaluate my performance, and ensure that my performance standards are changed if what I do at work changes</i> .
Seeking appeals ($\alpha = 0.86, 0.91$)	I have ways to <i>appeal a performance appraisal that I think is biased</i> . I can get a fair review of my performance appraisal if I ask for one and challenge a performance appraisal if I think it is unfair. My performance appraisal can be changed if I can show that it is incorrect or unfair. A process to appeal an appraisal is available to me anytime I may need it.
Ratings based on equity ($\alpha = 0.87, 0.93$)	The appraisal I get reflects <i>how much work I do, how well I do my work, the many things I do that help at work, the many things I am responsible for at work, and the effort I put forth at work</i> .
Ratings not based on politics ($\alpha = 0.68, 0.87$)	<i>My rater gives me the rating I earn even when it might upset me. My rating is not the result of my rater trying to avoid bad feelings among employees, higher than one I would earn based on my contribution to my organization or based on how much status I have. My rating is a result of my rater applying standards consistently across employees without pressure, corruption, or prejudice</i> .
Raters show respect ($\alpha = 0.91, 0.93$)	My rater is <i>rarely rude to me, almost always polite, and courteous to me; and my rater treats me with respect and dignity</i> .
Raters show sensitivity ($\alpha = 0.92, 0.93$)	My rater <i>does not invade my privacy, is sensitive to my feelings, treats me with kindness, shows concern for my rights as an employee, and does not make hurtful statements about me</i> .
Clarifying expectations ($\alpha = 0.87, 0.94$)	My rater explains to me <i>what he or she expects for my performance, the standards that will be used to evaluate my work and how I can improve my performance</i> . My rater gives me a chance to question how I should meet my work objectives and regularly explains to me what he or she expects of my performance.
Providing feedback ($\alpha = 0.93, 0.94$)	My rater <i>frequently lets me know how I am doing, gives me information I can use to improve my performance, routinely gives me feedback relevant to the things I do at work, reviews with me my progress towards my goals and lets me know how I can improve my performance</i> .
Explaining and justifying decisions ($\alpha = 0.88, 0.96$)	My rater <i>helps me to understand the process used to evaluate my performance, takes time to explain decisions that concern me, lets me ask him or her questions about my performance appraisal and gives me real examples to justify his or her appraisal of my work. My rater's explanations help to clarify for me what to do to improve my performance</i> .

Notes: Internal consistency estimates (α) reported for pilot study ($n = 45$) and total sample ($n = 188$). Each scale contains five items. Unique portions of items italicized

Table I.
Perceptions of fair
appraisal practices items

discrepancy score's positive mean ($M = 1.33$, $SD = 2.13$) indicated that, on average, the officers' actual performance ratings exceeded their beliefs about their successfulness, work quality and contribution toward the organization. Only 20 percent of the officers had a negative discrepancy score.

Phase 2. Confirming latent constructs

The purpose of the second phase of the process was to assess the underlying factor structure of the ten scales of fair appraisal practices perceptions. In order to confirm the factor structure, a nested analysis was accomplished to compare the relative fit of the four-factor model to two alternative three-factor models, two alternative two-factor models, and a single-factor model. Each of the alternative models was statistically over-identified, which allowed for the generation of model fit statistics. A combination of high reliability for each component scale, good model fit indices and a statistically significant improvement in fit over the alternative models provided evidence of the validity of the hypothesized component structure.

Participants included employees from four organizations ($n = 188$) operating under four different performance appraisal systems. Three of the organizations were from the United States Air Force. The fourth was a civilian organization. The first group of participants consisted of Air Force enlisted personnel ($n = 52$) from an Air Force Station in the western continental United States. These military employees were administrative specialists and hospital technicians working in skilled and semi-skilled positions with well-defined roles and tasks. All employees had received a performance appraisal within the previous twelve months. The second group of participants comprised Air Force officers ($n = 52$) stationed at an Air Force Base in the eastern continental United States. The officers were project managers, engineers, and lawyers working on fairly indeterminate tasks. All officers had received a performance appraisal within the previous twelve months. The third group that participated in the survey was comprised of employees ($n = 44$) from a fairly new division of a health insurance corporation in the mid-western United States. The employees primarily held administrative-support or customer-assistance positions. Tasks were well defined and routine. All employees had received a performance appraisal within the previous 30 days. The fourth group of participants consisted of civil service employees from an organization in the mid-western United States ($n = 40$) who were instructors of professional continuing education courses in acquisition management, logistics management, or civil engineering management. Individuals were responsible for course development and implementation. All employees had received a performance appraisal within the previous 60 days.

The data collected from the four organizations were analyzed in their entirety in order to assess the underlying factor structure of the ten scales of fair appraisal practices perceptions. Small samples prohibited separate analysis for each organization. Table II shows scale descriptive statistics, as well as the zero-order correlation coefficients and covariance coefficients among the ten scales of fair appraisal practices perceptions for the total sample. The nested comparison of competing structural models provided a test of the relationships of the ten performance appraisal practices scales to the underlying latent justice constructs. The hypothesized four-factor model was compared to several plausible alternative models in order to determine the factor structure that best described the covariance patterns in the data.

	<i>M</i>	Skew	Kurtosis	1	2	3	4	5	6	7	8	9	10
Assigning raters	5.8	-1.5	2.0	2.14	0.69	0.58	0.67	0.66	0.50	0.56	0.64	0.64	0.55
Setting criteria	5.0	-0.7	0.0	1.48	2.17	0.69	0.70	0.62	0.45	0.52	0.72	0.73	0.68
Seeking appeals	5.2	-0.6	-0.2	1.22	1.48	2.12	0.68	0.62	0.43	0.56	0.66	0.68	0.65
Equity norm	5.6	-1.3	1.2	1.42	1.49	1.43	2.10	0.78	0.64	0.69	0.75	0.76	0.71
Absence of politics	5.8	-1.1	0.6	1.17	1.11	1.09	1.37	1.45	0.61	0.70	0.78	0.77	0.73
Respect	6.3	-2.1	4.9	0.81	0.73	0.70	1.02	0.81	1.22	0.88	0.52	0.56	0.53
Sensitivity	6.2	-1.7	2.9	0.95	0.89	0.95	1.15	0.97	1.13	1.35	0.59	0.64	0.59
Clarifying	5.4	-1.0	0.3	1.41	1.59	1.44	1.63	1.41	0.87	1.03	2.24	0.90	0.90
Explaining	5.5	-1.0	0.4	1.36	1.55	1.44	1.61	1.36	0.90	1.08	1.95	2.11	0.90
Feedback	5.2	-0.9	0.2	1.30	1.61	1.51	1.66	1.40	0.95	1.10	2.17	2.10	2.57

Note: Scale mean (*M*) transformed by dividing by the number of items, covariances are shown below diagonal, variances shown on diagonal, and correlations shown above the diagonal in italics, *n*=188, all correlations are statistically reliable *p* < 0.01

Table II.
Descriptive statistics,
covariances and
zero-order correlations for
justice perceptions

The single factor model represents a generalized justice construct. The alternative two-factor models represent the traditional distinction between procedural and distributive justice and an alternative distinction between social and structural determinants of justice developed by Greenberg (1993). Because the two-factor models are nested in the single factor model, they can be directly compared to the generalized one-factor justice model. The two alternative three-factor models represent the current conflict in justice theory about the distinction between interactional and procedural forms of justice. One three-factor model distinguishes between distributive, interpersonal, and procedural dimensions of justice. This model is most similar to the arrangement of constructs in Moorman (1991) where the socially and structurally determined forms of procedural justice are combined into a single construct. This three-factor model is nested in the distributive-procedural model with the distributive construct split in two. The other three-factor model distinguishes the socially determined form of justice (interactional) from the two structural forms of justice (distributive and procedural). This is the model that is most similar to the arrangement of constructs in Skarlicki and Folger (1997) in which the two socially determined justice constructs (interpersonal and informational) are combined into a single construct. This three-factor model is nested in the two-factor social-structural model with the structural construct split in two.

The maximum likelihood estimation technique used in the LISREL (Jöreskog and Sörbom, 1993) structural equation model program assumes that the measured variables are continuous and have a multivariate normal distribution. Violations of these assumptions can result in overestimation of the χ^2 causing false rejections of true models, and can reduce standard error estimates that lead to increased chances of finding statistically reliable paths that are not true (West *et al.*, 1995). Monte Carlo studies have shown that maximum likelihood solutions are robust to skewness with only trivial effects on estimation of parameters and standard errors (Jaccard and Wan, 1996). The same studies, however, show that parameters and standard errors can be very sensitive to kurtosis. The skewness and kurtosis statistics for the ten justice scales in Table II indicate moderate departures from normality in several variables. The positive kurtosis in these variables can negatively bias the standard error estimates and create an increased chance of making a Type I error. Jaccard and Wan (1996) suggest using an estimation procedure other than maximum likelihood that is robust to departures from normality and West *et al.* (1995) suggest using a technique that adjusts the normal theory χ^2 and standard errors to account for the degree of multivariate kurtosis in the sample data. The EQS program (Bentler, 1997) has a robust maximum likelihood solution that provides standard errors that have been corrected for non-normal data. The maximum likelihood statistics are divided by a constant k based on the model implied residual matrix, the observed multivariate kurtosis, and the model degrees of freedom (Santorra, 1990). This approach appeared reasonable for the moderate departure from normality in the total sample.

Phase 3. Assessing justice perceptions and hypothesized consequences

The purpose of the third phase was to investigate the relationships between the underlying latent justice dimensions and their hypothesized consequences[1]. Competing models of the nomological network were analyzed using the EQS (Bentler, 1997) structural equation modeling program. The nested comparison of

competing structural models provided a direct test of the fourth hypothesis concerning the relationships of justice perceptions to employees' reactions toward their appraisal, appraisal system, supervisor and organization. Investigation of the direct and indirect relationships between the four justice dimensions and their anticipated consequences provided tests of the remaining hypotheses.

Employee reactions measures. In addition to the ten justice scales of perceptions of performance appraisal practices, the revised Employee Survey included measures of perceptions of their own work performance, perceptions of their most recent performance appraisal, and measures of satisfaction with their appraisal, appraisal system, and supervisor. Both component measures for the discrepancy perceptions construct had good internal consistency ($\alpha = 0.85$ for Perceptions of Work Performance and $\alpha = 0.93$ for Perceptions of the Performance Appraisal). The internal consistency estimate for the discrepancy score ($\alpha = 0.84$) was also satisfactory. The reliability index for the discrepancy score as a function of the component reliabilities and correlation between the two components ($r_{xy} = 0.60$) was lower ($r_{xx} = 0.73$), but still above acceptable minimum standards. About 27 percent of the sample had negative discrepancy scores and 37 percent had positive discrepancy scores. On average, the positive discrepancy was due to the third item, "contributions to the organization". Perceptions of "work performance" and work quality" had slightly negative average difference scores. Employee's reactions concerning their most recent performance appraisal, the performance appraisal system, and their supervisor were measured using items modified from previous studies (e.g. Tang and Sarsfield-Baldwin, 1996; Taylor *et al.*, 1995). These measures used seven-point Likert scales, ranging from "Strongly disagree" (1) to "Strongly agree" (7). Single item measures for the satisfaction constructs were chosen that did not suffer from content overlap with the justice constructs which may have lead to spurious correlations between the measured variables (Stone-Romero, 1994). Satisfaction with the performance appraisal system was measured with the item "I am satisfied with the system used to evaluate my performance". Satisfaction with the most recent performance appraisal was measured with the item "I am satisfied with the performance appraisal I received this last rating period". Satisfaction with the supervisor was measured with the item "All in all, I think I have a good supervisor".

Supervisor measures. A second survey collected supervisor's reports of pro-social work behaviors exhibited by their employees. Ten items based on Williams and Anderson's (1991) scales were used to measure organizational citizenship behaviors beneficial to the supervisor and organizational citizenship behaviors beneficial to the organization ($\alpha = 0.92$). Supervisors were asked how likely it was for their employee to engage in behaviors they found helpful. The Citizenship Behaviors toward the Supervisor scale included items to measure the extent that employees helped out the supervisor, passed on work related information, made positive comments about the supervisor, sacrificed personal time to help the supervisor, and went out of the way to provide helpful information. Supervisors were also asked how likely it was for the employee to engage in behaviors that were helpful to the organization ($\alpha = 0.86$). The Citizenship Behaviors toward the Organization scale included items to measure the extent that employees tackled difficult work enthusiastically, conserved and protected organizational property, said positive things about the organization, gave advance notice when unable to come to work, and adhered to informal rules devised to maintain order.

Combined sample. Participants comprised a subset of those employees having matched surveys with their immediate supervisors ($n = 117$). All four organizations were represented in the matched sample ($n = 18$ for the enlisted personnel, $n = 33$ for the officers, $n = 26$ for the health insurance employees, and $n = 40$ for the civil servants). The means, standard deviations, skewness, kurtosis, and error variances, as well as zero-order correlation coefficients and covariance coefficients among the ten scales of fair appraisal practices perceptions, discrepancy, single item satisfaction measures and supervisors' reports of citizenship behaviors for the matched sample are depicted in Table III.

Analyses. Competing structural equation models were analyzed to test the hypothesized relationships of justice perceptions with employee reactions as well as provide evidence for the construct validity of the four justice dimensions. The discrepancy and justice models could not be directly compared because one was not nested in the other. The models could however, be compared to a combined model that included all paths in both the discrepancy and justice models. The EQS (Bentler, 1997) structural equation modeling programs provide unreliability estimates for measured variables when multiple indicators are chosen for latent constructs (e.g. the ten justice perception scales for the four latent justice constructs). EQS cannot estimate the unreliability of a measured variable when each latent construct is represented by a single measured variable. Unreliability estimates for single indicators must be provided. To account for the unreliability of the discrepancy, attitude, and behavior measures, the theta delta and theta epsilon matrices were constrained to predetermined levels of error variance corresponding to each measure's estimate of unreliability. To calculate the error variance, the scale reliability estimate was subtracted from 1.0 and then multiplied by the scale variance ($e = (1 - r_{xx}) * s^2$). Reliability estimates for the discrepancy difference score, and the supervisors' reports of citizenship behaviors were based on α . Internal consistency reliability estimates could not be calculated for the single item satisfaction measures. Two estimates for reliability (0.85 and 0.95) were used for the single item measures. Path coefficients, correlations among latent constructs, and residual errors are reported using both estimates of reliability for the final models.

The justice and alternative discrepancy models were compared to null models in which no paths were specified between perceptions and hypothesized consequences and a combined model that contained all paths specified in the justice and discrepancy models. Four conditions needed to be met in order for the hypothesized model to be considered the best fitting model. First, the justice model had to provide a statistically reliable improvement in fit over the null model. Second, comparison of the justice model to the combined model should not provide a statistically reliable improvement in fit. Third, the combined model had to provide a statistically reliable improvement in fit over the discrepancy model. Fourth, the justice model should have good model fit scales, and no evidence of ill fit.

Results

Confirming latent constructs

The results of the nested confirmatory factor analyses (Table IV) imply that the hypothesized four-factor model (F) provides a better explanation of the underlying patterns in the ten scales than the other models. The statistically reliable model χ^2 for the four-factor model suggests that the specified paths did not provide a perfect fit to the data. Jaccard and Wan (1996) describe three additional classes of fit scales

	<i>M</i>	Skew	Kurtosis	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Assigning raters	6.11	-2.0	4.5	1.96	0.58	0.43	0.62	0.56	0.51	0.52	0.54	0.61	0.50	0.35	0.50	0.49	0.59	0.22	0.26
Setting criteria	5.3	-0.7	0.0	0.92	2.25	0.63	0.63	0.55	0.43	0.47	0.68	0.72	0.67	0.21	0.62	0.48	0.54	0.20	0.26
Seeking appeals	5.3	-0.4	-0.5	0.67	1.12	2.25	0.58	0.50	0.36	0.47	0.55	0.58	0.60	0.17	0.56	0.48	0.50	0.11	0.12
Equity norm	5.8	-1.5	2.0	1.03	1.16	1.05	2.52	0.74	0.63	0.65	0.70	0.74	0.70	0.36	0.70	0.66	0.70	0.24	0.27
Absence of politics	5.9	-1.0	0.4	0.76	0.83	0.74	1.16	1.88	0.57	0.61	0.76	0.77	0.74	0.23	0.57	0.51	0.72	0.14	0.20
Respect	6.4	-2.3	5.4	0.67	0.63	0.53	0.97	0.71	1.92	0.89	0.48	0.55	0.51	0.43	0.53	0.66	0.64	0.11	0.07
Sensitivity	6.3	-1.7	1.6	0.68	0.69	0.68	0.99	0.75	1.07	1.91	0.51	0.61	0.54	0.42	0.51	0.65	0.72	0.15	0.16
Clarifying	5.5	-1.0	0.6	0.90	1.27	1.02	1.35	1.20	0.74	0.77	2.42	0.87	0.91	0.26	0.51	0.52	0.70	0.15	0.25
Explaining	5.5	-1.0	0.8	1.00	1.31	1.05	1.40	1.19	0.83	0.91	1.65	2.39	0.90	0.38	0.52	0.55	0.72	0.17	0.23
Feedback	5.2	-0.9	0.2	0.94	1.41	1.25	1.53	1.32	0.89	0.93	2.01	1.94	2.93	0.32	0.48	0.53	0.69	0.16	0.23
Difference score	0.1	-0.8	4.4	0.32	0.20	0.17	0.38	0.20	0.35	0.35	0.27	0.39	0.37	1.15	0.24	0.42	0.29	0.18	0.20
Systems satisfaction	5.5	-1.1	0.2	1.05	1.47	1.32	1.72	1.14	1.04	0.99	1.28	1.25	1.35	0.32	3.62	0.58	0.57	0.18	0.16
Appraisal satisfaction	6.4	-2.4	6.1	0.69	0.76	0.75	1.07	0.68	0.86	0.83	0.85	0.87	0.99	0.37	1.21	2.19	0.55	0.18	0.23
Supervisor satisfaction	6.0	-1.7	2.0	1.09	1.12	1.03	1.51	1.26	1.09	1.21	1.51	1.53	1.69	0.33	1.57	1.00	3.06	0.15	0.22
OCB to organization	6.0	-0.9	0.3	0.21	0.22	0.11	0.28	0.13	0.09	0.14	0.18	0.18	0.20	0.11	0.26	0.18	0.19	1.34	0.80
OCB to supervisor	5.8	-0.8	0.4	0.30	0.33	0.15	0.36	0.21	0.07	0.16	0.33	0.30	0.34	0.14	0.27	0.26	0.33	0.62	1.40

Note: Scale mean (*M*) transformed by dividing by the number of items, covariances are shown below diagonal, variances shown on diagonal, and correlations shown above the diagonal in italics, *n* = 117, correlations greater than 0.15 are statistically reliable *p* < 0.05

Table III.
Descriptive statistics,
covariances and
zero-order correlations for
perceptions, attitudes and
behaviors – matched
sample

Table IV.
Comparison of nested
models of justice
perceptions for the total
sample

Model	$\chi^2(df)$	GFI	$\chi^2\text{diff}(df)$	ΔGFI
A. Justice	372.8* (35)	0.71		
B. Procedural-distributive Difference between model A and model B	278.6* (34)	0.78	94.2* (1)	0.07
C. Procedural-distributive-interpersonal Difference between model B and model C	133.2* (32)	0.86	145.4* (2)	0.08
D. Structural-social Difference between model A and model D	315.8* (34)	0.78	57.0* (1)	0.07
E. Procedural-distributive-interactive Difference between model D and model E	300.4* (32)	0.78	15.4* (2)	0.00
F. Procedural-distributive-interpersonal-informational Difference between model C and model F Difference between model E and model F	67.7* (29)	0.94	65.5* (3) 232.7* (3)	0.08 0.16

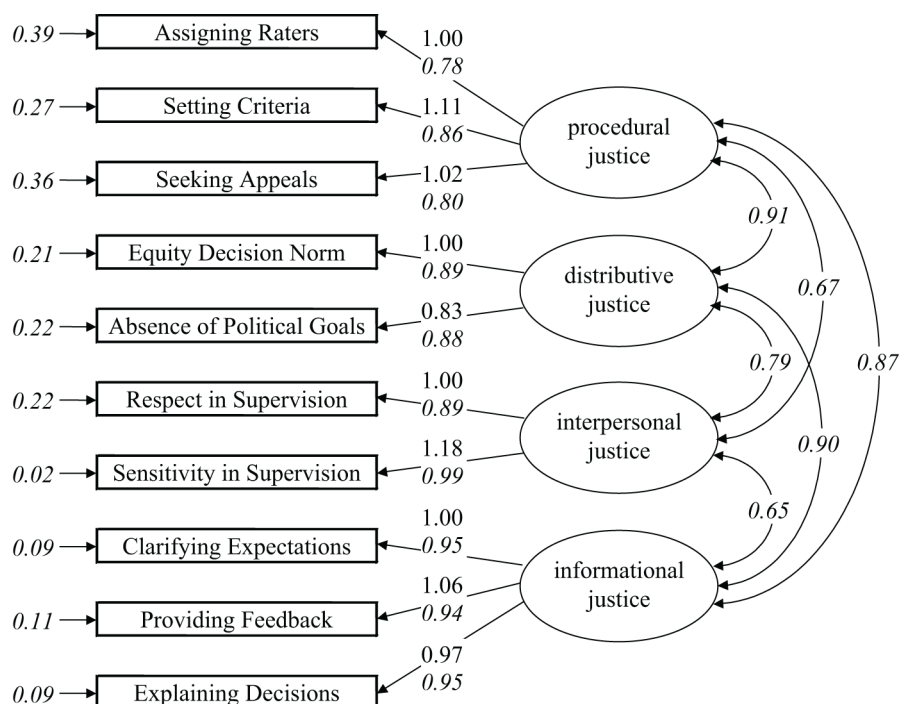
Note: $n = 188$; Goodness of fit index (GFI), * $p < 0.05$, χ^2 degrees of freedom for each model and difference between models given in parentheses

(absolute, parsimonious, and relative) that should be considered when evaluating the fit of a structural equation model. Absolute fit compares the predicted and observed covariance matrices. Both the goodness of fit index (GFI = 0.94) and standardized root mean square residual (Standardized RMR = 0.026) indicated satisfactory absolute fit to the model. The standardized RMR of 0.026 indicated that the average deviation between the predicted and observed correlations was less than the recommended threshold of 0.05. The second category of fit scales also considers absolute fit, but penalizes the model based on its complexity. The more paths specified, the lower the models' parsimony. The root mean square error of approximation (RMSEA) of 0.084 is close to the acceptable threshold of 0.08 for adequate parsimonious fit. The third category of fit scales compares the absolute fit to an alternative model. The value for the comparative fit index (CFI = 0.98) indicates that the four-factor model has a good fit compared to a null model that posits no correlations between the observed variables.

The EQS robust standardized and unstandardized path coefficients for the ten scales of perceptions of fair appraisal practices and correlations among the four latent constructs are depicted in Figure 1. Overall, the results indicated that the underlying structure of the ten scales of appraisal practices perceptions was consistent with the four dimensions of justice theorized by Greenberg (1993). The evidence suggested that the four justice categories (procedural, distributive, interpersonal and informational) were distinct but highly correlated constructs, consistent with Colquitt (2001). The four-factor model provided relatively small but statistically reliable improvements (ΔGFI of 0.08) over the three-factor model that combined the procedural and informational justice dimensions into a single procedural justice construct.

Assessing justice perceptions and hypothesized consequences

The results of the nested analyses of the perceptions, attitudes, and supervisors' reports of organizational citizenship behaviors for the matched sample of 117 employees and supervisors across the four different organizations appear in Table V.



Notes: $n = 188$, $\chi^2(29) = 67.9$, CFI = 0.98, Robust CFI = 0.99. All paths are statistically reliable at $p < 0.001$, standardized paths appear in italics

Figure 1.
Confirmatory factor
structure.

Model		$\chi^2(df)$	GFI	$\chi^2 \text{diff}(df)$	ΔGFI
A.	Null	365.5* (96)	0.78		
B.	Discrepancy	320.2* (89)	0.79		
C.	Justice	169.7* (89)	0.86	45.3* (7)	0.01
D.	Combined	164.7* (85)	0.86	150.4* (0)	0.07
	Difference between model B and model D			200.8* (11)	0.08
	Difference between model C and model D			5.0 (4)	0.00

Note: Subgroup $n = 117$; Goodness of fit index (GFI), * $p < 0.05$, χ^2 degrees of freedom for each model and difference between models given in parentheses

Table V.
Comparisons for
structural equation
models of perceptions,
attitudes and behaviors

The hypothesized justice model (C) provided a statistically reliable improvement over the null model (A), and no statistically reliable difference from the combined model (D). The combined model (D) did, however, offer a substantial improvement in fit over the discrepancy model (B) indicating that the discrepancy model omitted important relationships among constructs. The justice model and the more constrained combined model had statistically equivalent fit. By rule of parsimony, the simpler of the two

justice models was accepted as the better fitting model. These results provide evidence supporting the first three conditions for concluding the hypothesized justice model (C) as best fitting. The statistically reliable χ^2 for the justice model suggests that the specified paths did not provide a perfect fit to the data. Only one index pointed to satisfactory fit for the matched sample. The GFI of 0.86 indicated less than satisfactory absolute fit. The standardized RMR of .06 indicated that the average deviation between the predicted and observed correlations exceeded the accepted threshold of 0.05. The RMSEA of 0.088 exceeded the acceptable threshold of 0.08 indicating adequate fit for parsimony. The CFI value of 0.95 indicated that the justice model had a good fit compared to a null model that posits no relationships between the justice, discrepancy, and satisfaction variables. An inspection of the fitted and standardized residuals revealed 29 statistically reliable residuals ($z > 1.96, p < 0.05$). The proportion of statistically reliable residuals (29/136) exceeded the guideline of five percent. An investigation of the modification indices revealed no theoretically reasonable changes to improve the model fit.

The EQS robust standardized coefficients for the paths between latent variables are depicted in Figure 2. The paths are given in the same figure for solutions where the single item satisfaction reliability estimates were set at 0.85 and 0.95. Changing the reliability estimates had no effect on the fit statistics for the model. The different reliability estimates did, however, change the standardized path coefficients and the estimates of unexplained variance in the latent endogenous constructs. Standardized path coefficients were larger and estimates of unexplained variance were lower when the reliabilities of the satisfaction scales were set as 0.85 compared to 0.95. All paths from the latent exogenous justice perception constructs to the latent endogenous satisfaction constructs were statistically reliable. The path from the discrepancy construct to satisfaction with the current appraisal, however, was not statistically

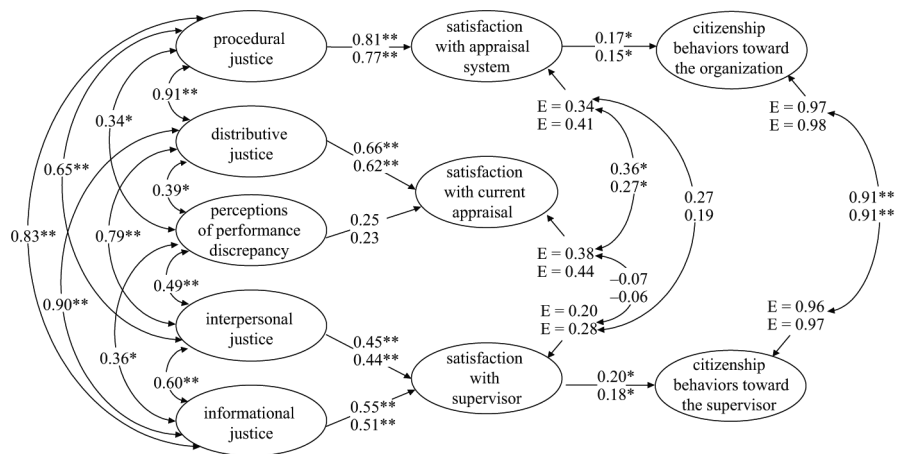


Figure 2.
Hypothesized justice
model of perceptions,
attitudes and behaviors
for the matched sample

Notes: Standardized robust estimators shown, * $p < 0.05$; ** $p < 0.01$. Top coefficient is for satisfaction reliability set to 0.85, bottom coefficient for reliability set at 0.95. Latent residual terms correlated to account for common sources of variance other than what is included in the model. Sample $n = 117$, $\chi^2 (89) = 169.5$, CFI = 0.95, Robust CFI = 0.96

reliable. The paths from the satisfaction constructs to their hypothesized consequences were also statistically reliable. Affective reactions to the performance appraisal system and to the employee's supervisor were predicted by justice perceptions, not by the discrepancy between perceptions of work contributions and the received performance appraisal. This provided support for *H1*.

The different justice dimensions also correlated with different consequences in predictable ways providing support for *H2-H4*. Perceptions of distributive justice were related to satisfaction with the performance appraisal. Modification indices indicated no potential improvements to model fit for adding consequences to the distributive justice construct, and no improvement in fit by adding paths from the procedural, interpersonal or informational justice dimensions to the appraisal satisfaction construct. The evidence suggests that even though the procedural, informational, and distributive justice constructs are highly correlated, distributive justice is a better predictor of employees' satisfaction with their current performance appraisal than the other justice dimensions. Perceptions of procedural justice were related to satisfaction with the performance appraisal system. Modification indices indicated no potential improvements to model fit for including additional consequences to the procedural justice construct, or for adding paths from the discrepancy, distributive or interpersonal perceptions to the appraisal system satisfaction construct. The modification indices in the gamma matrix, however, suggested creating a path from informational justice to satisfaction with the performance appraisal system. Including this path in the model produced a theoretically meaningless negative coefficient, and a negligible improvement in fit ($\Delta\text{GFI} = 0.00$). Substituting the information justice construct for the procedural justice construct produced a theoretically meaningful and statistically reliable positive path from information justice to appraisal system satisfaction, but produced a decrement in model fit ($\chi^2_{\text{diff}} = 57.0$, $p < 0.05$, $\Delta\text{GFI} = 0.04$) and a reduction in the amount of variance accounted for in appraisal system satisfaction. The evidence suggests that even though the procedural and informational justice constructs are highly correlated, procedural justice is a better predictor of employees' satisfaction with the performance appraisal system than the informational justice type. Perceptions of interpersonal and informational justice were both positively related to satisfaction with the supervisor. Modification indices indicated no potential improvements to model fit for adding consequences to either of the interpersonal or informational justice constructs; and no improvement in fit by adding paths from the discrepancy, procedural or distributive constructs to supervisor satisfaction. The evidence suggests that both of the socially determined justice constructs are important predictors of employees' satisfaction with their supervisor.

The results also indicate that the justice perceptions were indirectly related to behavioral consequences in predictable ways, providing support for *H5-H6*. Procedural justice perceptions were indirectly related to supervisor's reports of helpful behaviors toward the organization through satisfaction with the appraisal system. The relationship was statistically reliable ($p < 0.05$). Interpersonal and informational justice perceptions were indirectly related to reports of helpful behaviors toward the supervisor through satisfaction with the supervisor. These relationships were also statistically reliable ($p < 0.01$). Although the relationships were statistically reliable, the effect sizes were small. The justice perceptions and attitudes explained between two and three percent of the variance in supervisors reports of helpful

behaviors toward the organization, and between three and four percent of the variance in the reports of helpful behaviors toward the supervisor. The range in the proportions of explained variance depends on whether the reliability for the satisfaction variables was set at 0.95 (lower estimate) or 0.85 (higher estimate).

Discussion and conclusions

Perceptions of inaccuracy and injustice as well as feelings of dissatisfaction have long plagued performance appraisals and the organizational processes that generate them. Folger *et al.* (1992) suggested the possibility that researchers and practitioners have not been able to substantially change employees' affective reactions about performance appraisal systems because their efforts have been embedded in an overly rational conceptualization. They suggested augmenting the traditional rational approach that viewed performance appraisals as equivalent to test construction, with an approach based on organizational justice. In order to assess this alternative approach, a set of scales was developed based on organizational justice theory to measure employees' perceptions of their organizations' performance appraisal practices. The results of this study reaffirmed the underlying structure of justice perceptions, and provided an empirical test of Folger *et al.*'s argument for an organizational justice approach to understanding performance appraisal practices.

If organizational leaders have a reasonably complete understanding of employees' perceptions about the performance appraisal system and process, they can modify their performance appraisal practices so that their employees believe the systems and processes are informative and fair. Ten multi-item scales, based on the seminal works in the organizational justice literature, were designed to measure individuals' perceptions of the extent to which fair processes and interactions are manifest in their organization's performance appraisal systems. Content validity, item clarity, and conceptual distinction among the four justice categories were ensured through a confirmatory categorization analysis using subject matter experts. The pilot and subsequent study using these scales demonstrated good psychometric properties for scales that measured employees' perceptions of the processes associated with assigning raters, setting criteria, seeking appeals, clarifying expectations, explaining decisions, and providing feedback; as well as their perceptions of the outcomes associated with the equity rule, absence of political goals, and respectful and sincere treatment. These ten scales were adequate to provide an empirical evaluation of the underlying dimensions of organizational justice.

The analysis revealed that the four justice dimensions (procedural, distributive, interpersonal, informational) are distinct, but highly correlated constructs, which supports Colquitt's (2001) assertions to consider the four justice dimensions separately. However, it is important to point out the high correlations among the justice scales and only minor improvements in incremental fit; thus the practical importance of making distinctions between these constructs remains questionable.

If justice perceptions of the procedures and social interactions are important to employees, then these perceptions should influence attitudinal and behavioral reactions beyond the effects of the initial discrepancy between expected and actual performance ratings. A justice model which differentiated the effects of the four justice facets on affective reactions and supervisor reports of behaviors provided a better fit to the data than a performance discrepancy model based on the traditional rational

approach to performance appraisal systems. Consistent with referent cognition theory (Cropanzano and Folger, 1989), discrepancy perceptions predicted employees' attitudes toward their performance appraisal, while attitudes toward the system and supervisor were predicted by the employees' perceptions of the fairness of the performance appraisal practices. The various types of justice perceptions indeed predicted different affective consequences in the performance appraisal context. Four paths were specified between latent justice constructs and their hypothesized consequences. All four of these paths were statistically reliable providing support for the second, third, and fourth hypotheses.

Evidence from the competing structural equation models of perceptions and attitudes combined with concurrent validity evidence of relationships between specific justice dimensions and hypothesized consequences bolstered the argument for the distinction of four justice dimensions. Addition of the supervisors' reports of subordinate behaviors to the competing models of the nomological network provided another opportunity to address the practical importance of distinguishing between justice dimensions. The statistically reliable indirect relationships between the structural procedural justice type and citizenship behaviors toward the organization, and between the socially determined interpersonal and informational justice dimensions and citizenship towards the supervisor provided empirical support for *H5-H6*. The predictive power of these constructs on supervisor reports of employee citizenship behaviors, however, appears marginal. Also, it is important to note that even though interpersonal and informational justice are viewed as distinct, more research is needed to test whether these justice dimensions predict different outcome variables.

Limitations and suggestions for further research

Overall, this study makes several important contributions, but some limitations do exist. First, a sincere attempt was made in the development of the justice measures of performance appraisal practices to avoid the mono-operation bias that has plagued organizational justice measures in the past (Greenberg, 1990), but some bias is still present. Even though several scales were developed using multiple operational definitions, and perspectives were gathered from both employees and their supervisors for OCBs, all data were collected through a paper and pencil survey, which raises concern for common method bias. In the future, researchers should attempt to capture other responses, besides that of the supervisor, to better assess employees' actions that are supportive of organizational goals. In addition, different techniques for eliciting employee perceptions and attitudes, such as interviews or critical incidents should be used to expand the quantity and quality of information collected.

Second, the timing of the collection of data for the items that comprised the performance appraisal discrepancy construct is a concern. All responses from the employees were collected using a single instrument that was administered some time after the employees received their performance appraisals. The delay was less than one year for the Air Force officers and enlisted members, up to two months for the Air Force civil servants, and about one month for the employees from the health insurance organization. If the tenets of Cropanzano and Folger's (1989) referent cognitions theory are correct, the employees' beliefs likely changed during the time that elapsed prior to the collection of their perceptions and affective reactions. The high positive kurtosis for

the discrepancy data suggests that this may have occurred. More than one third of the employees (45 of 117) had no discrepancy, and nearly two thirds of the employees had a single point discrepancy on only one of the three items (discrepancy score of -0.33 to $+0.33$). In the future, researchers should collect the self-perceptions of performance during the last rating period before the employees receive the actual performance appraisal. After the employees have time to review and discuss their appraisals with their raters, data could then be collected on their perceptions of the appraisal, appraisal practices, and affective responses. At the conclusion of the survey, the employees could be asked to complete a second set of items concerning their expectations which could then be compared to those collected before the appraisal was received.

Third, conclusions cannot be drawn about causality given the non-experimental design of the study. Thus, reverse causality cannot be ruled out. For instance, employee attitudes about their organization or supervisor may influence their interactions with their supervisor as well as their individual performance, and in turn, employees could then justify their perceptions to be consistent with their attitudes. Future research should test these relationships using a true or quasi-experimental design (see Taylor *et al.* (1995).

Finally, this study was conducted in four organizations which were chosen for their convenience. The samples should in no way be considered representative of officer, enlisted and civilian personnel in the Air Force as a whole, nor should the employees of the health insurance organization be considered representative of any larger population. More work is needed to see if these results generalize to other organizations or industries, especially in cases where individuals feel that the performance appraisal system is not meeting their expectations. We recommend that the scales developed in this study be used to replicate and extend these finding in future studies.

Practical and theoretical importance of research

This research is important to theory for several reasons. First, this study reaffirms the four factor structure of organizational justice (Colquitt, 2001) and then integrates these justice dimensions to a specific context: performance appraisal. Second, our results indicate that justice dimensions differentially predict affective responses toward the appraisal, the appraisal system, and the supervisor, consistent with Sweeney and McFarlin's (1993) two-factor model and Bies and Moag's (1986) agent-system model. Third, this study supports the notion that justice dimensions influence behavioral responses (i.e. supervisors' reports of two types of organizational citizenship behaviors) through attitudinal responses. Finally, this study applied justice constructs to measurable organizational phenomena. Where previous studies (e.g. Colquitt, 2001; Roch and Shanock, 2006) have developed items to measure the justice constructs directly, this study developed scales that measure specific performance appraisal practices and then mapped them to the justice framework. We recommend this type of approach be used in the future when applying justice dimensions to organizational practices for employee selection and hiring, promotion, rewards, layoffs, and conflict resolution.

This research also has important practical applications. This study provides leaders with a set of measurement tools to gauge their performance appraisal systems. Organizational leaders can use these scales to determine where their performance appraisal systems seem unfair and then use this information to make sensible

decisions concerning their existing performance appraisal systems. The scales can be used by Human Resources professionals to uncover potential weaknesses in the existing performance appraisal system and to develop interventions to address these issues. For example, if employees perceive that raters are not showing respect and sensitivity, then organizations may wish to offer rater training to allow managers practice at conveying performance information in a sensitive manner. The scales can also be directly used by supervisors to assess their personal performance appraisal practices. Supervisors may not be able to influence the larger performance appraisal system in the short-term, but they can make changes to the way they provide information, collect and weigh evidence, and communicate progress.

One final thought. Cropanzano *et al.* (2007, p. 34) make a strong case that organizational justice can “create powerful benefits for organizations and employees” including “greater trust and commitment, improved job performance, more helpful citizenship behaviors, improved customer satisfaction, and diminished conflict”. Organizational activities like performance appraisal are necessary, but also risk the “ill will of employees”. Just practices have the potential to allow managers to make “tough decisions more smoothly” (Cropanzano *et al.*, 2007, p. 45). The results of this study provide evidence of this effect – employees’ perceptions of fairness of performance appraisal practices are linked to organizationally relevant attitudes and behaviors; beyond the influence of the discrepancy between expected and actual performance ratings. Organizational performance appraisal decisions may initially seem unfair, but they don’t have to be unfair.

Note

1. Separate subgroup analysis was conducted for professionals (Air Force Officers and Department of Defense civilians) and paraprofessionals (Air Force Enlisted and administrative support and customer assistance health insurance employees) with similar results as the full sample.

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Further reading

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